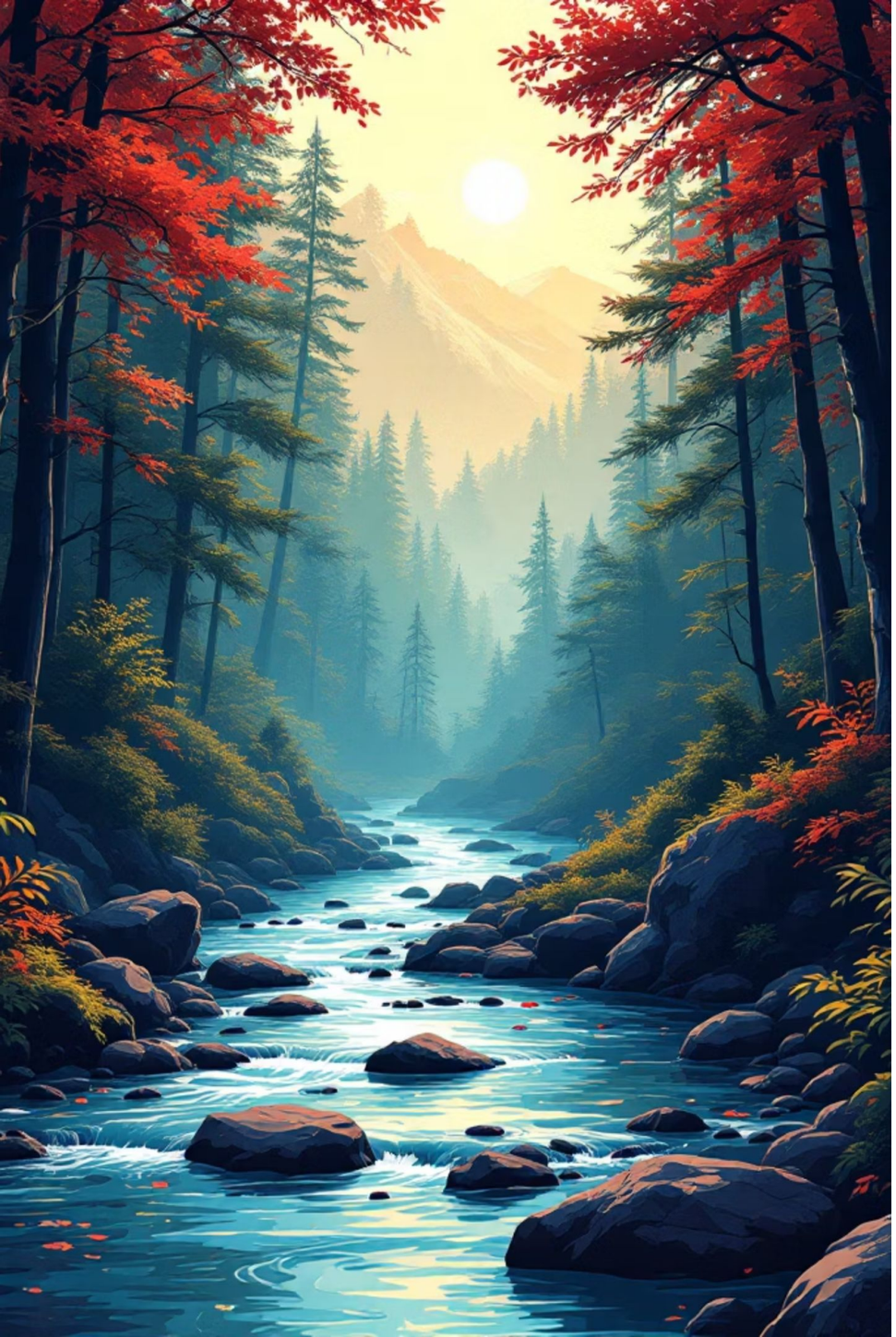


AI for Communication and Marketing

Mar 11, 2026 - Lab lecture #1: Introduction

Giulia Pennacchioni | giulia.pennacchioni@gmail.com

Head of Customer Analytics, QVC Italy

Assistant Professor

Course structure & final assessment

Weekly Practice Labs

Hands-on group sessions: using Gen AI & Agentic AI to solve real world problems

Final assessment

Individual project: end-to-end evaluation, analysis and building of models for marketing use cases

Grading breakdown: the individual project is worth 50% of the final exam. Some lessons will be dedicated to the development of the individual project.

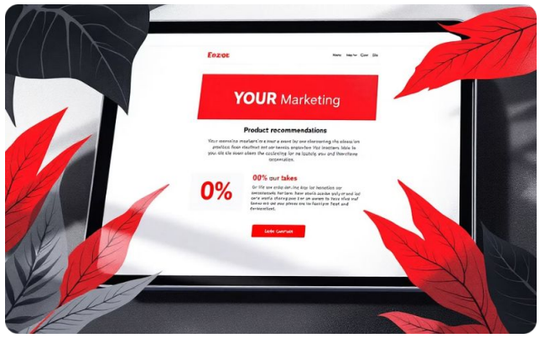
Subject: **We miss you! 15€ discount for you**

Subject: **Make yourself a gift: 10€ discount**

AI for Personalized Re-engagement

Behavioral Analysis

AI identifies lapsed customers by analyzing purchase history and engagement patterns.



Tailored Communications

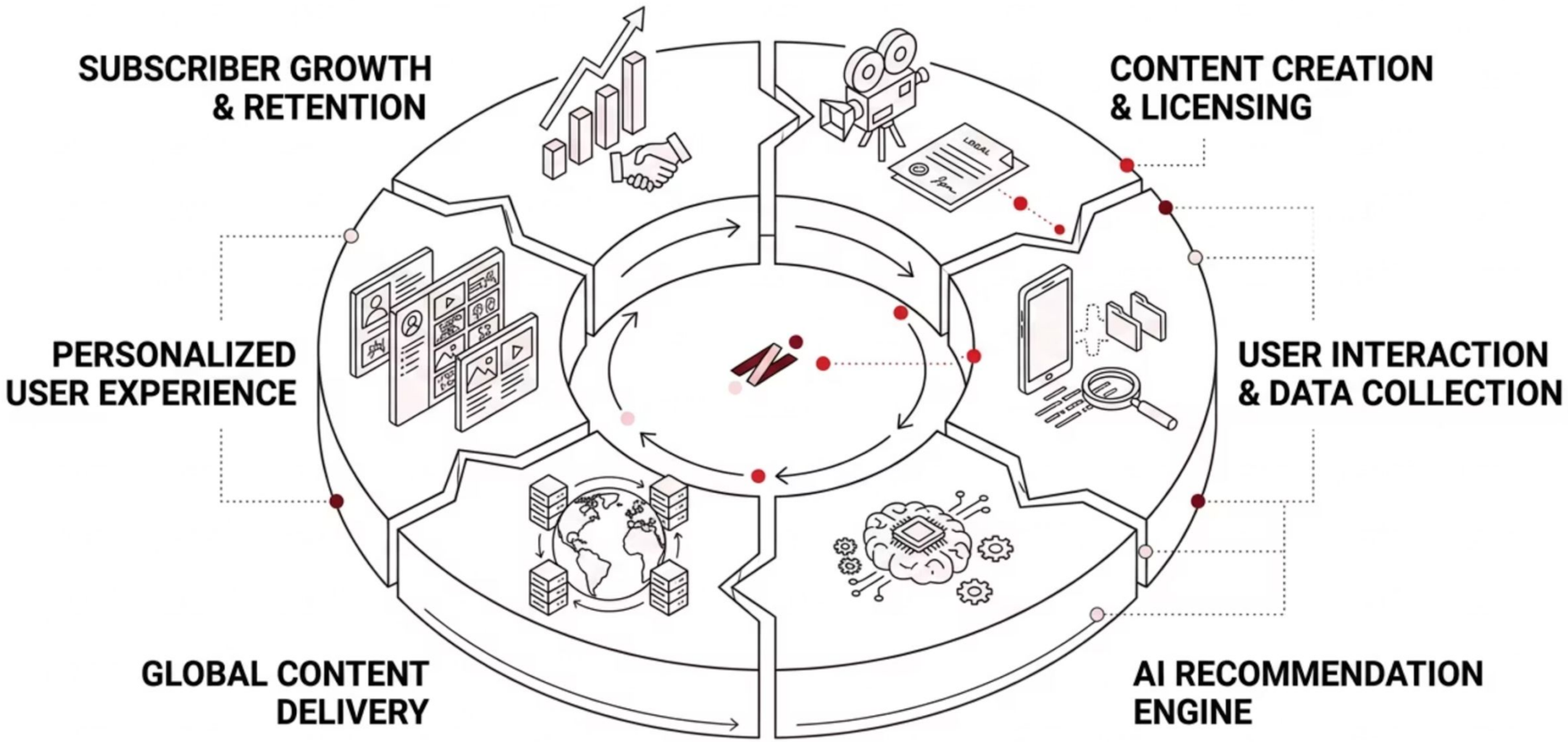
Personalized emails and offers are crafted based on unique past interests and preferences.

Reactivation Success

Increased return rates and customer lifetime value through targeted incentives.

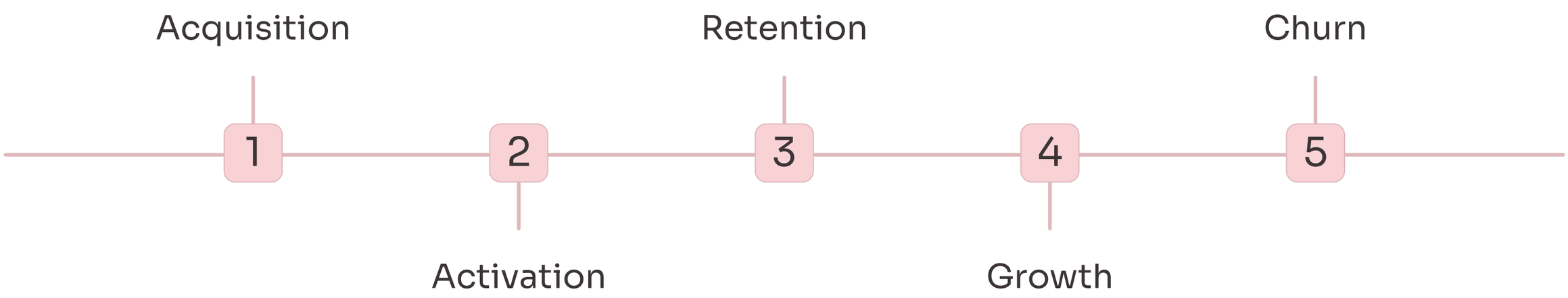
The AI-Powered Entertainment Engine

Netflix leverages a sophisticated AI ecosystem to personalize content, optimize delivery, and drive continuous engagement for its global audience.



This continuous cycle of content, data, and AI-driven personalization ensures a tailored experience, keeping subscribers engaged and attracting new ones.

Customer Lifecycle



AI Roles in Marketing Teams

Data Analysts

Transform raw data into actionable insights by identifying trends and patterns through statistical analysis

Data Scientists

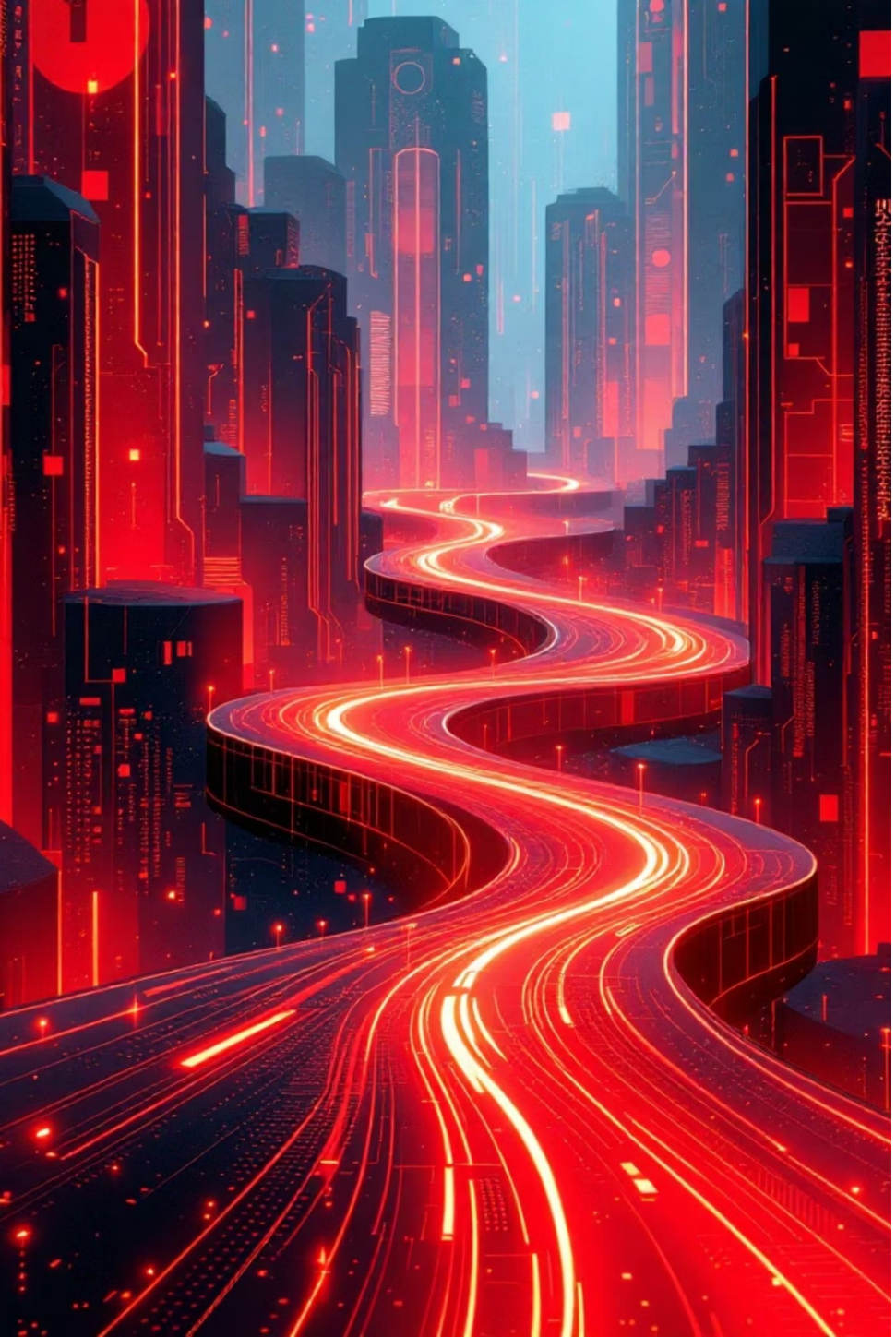
Build predictive models, design experiments, and analyze campaign effectiveness using A/B testing frameworks

Data Engineers

Deploy models to production, build data pipelines, and optimize inference latency for real-time personalization

Business Marketers

Leverage model outputs for targeting, create automated workflows, and measure ROI across channels



What Do You Want to Be?

Essential Tools for AI & Data Science

Git

The standard version control system for tracking code changes and enabling team collaboration.

IDE

The primary coding environment; VS Code or compatible editors with notebook extension integrated with AI agents.

Conda & Mamba

Package and environment managers; Mamba is the faster, C++ based alternative to Conda for handling dependencies.

Kaggle

The premier platform for discovering high-quality datasets, participating in competitions, and sharing notebooks.

The Python Ecosystem for Data Science

Python stands as the versatile, high-level programming language that forms the indispensable backbone for advanced AI and Data Science applications, powered by a rich ecosystem of specialized libraries.

1

NumPy

The fundamental package for scientific computing, providing support for powerful N-dimensional arrays and essential mathematical functions.

Pandas

The essential tool for efficient data manipulation and analysis, offering high-performance data structures, most notably DataFrames, for structured data.

Scikit-Learn

The go-to library for accessible and efficient machine learning, featuring comprehensive tools for classification, regression, clustering, and more.

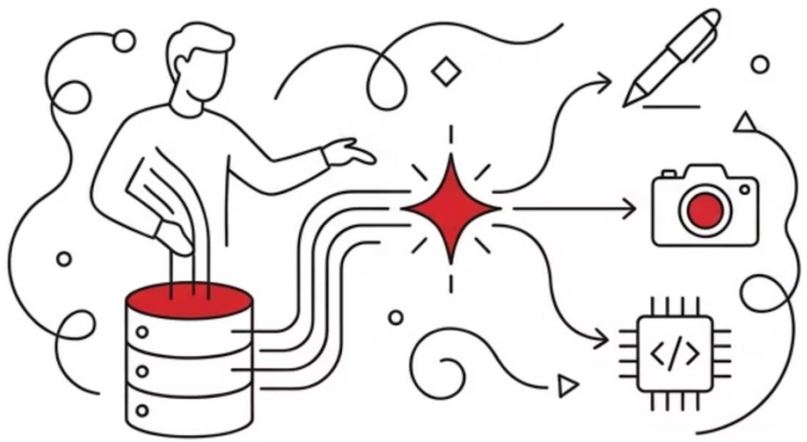
Matplotlib & Seaborn

The standard visualization toolkit; Matplotlib offers precise control over plots, while Seaborn provides beautiful, high-level statistical graphics for insightful data exploration.

Generative AI vs. Agentic AI

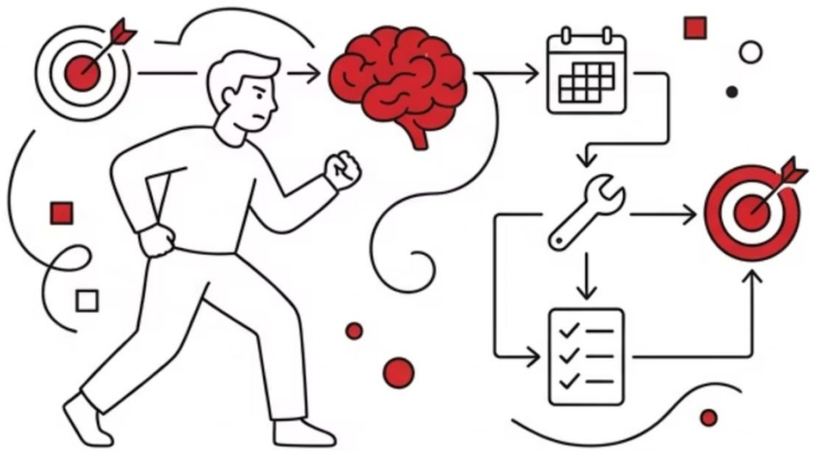
While both are advanced forms of artificial intelligence, Generative AI and Agentic AI serve distinct purposes, fundamentally differing in their primary function and operational approach.

GENERATIVE AI



- Creates new content: text, images, code.
- Focuses on CREATIVITY & SYNTHESIS.

AGENTIC AI



- Acts autonomously on GOALS.
- Focuses on PLANNING & MULTI-STEP PROBLEM SOLVING.

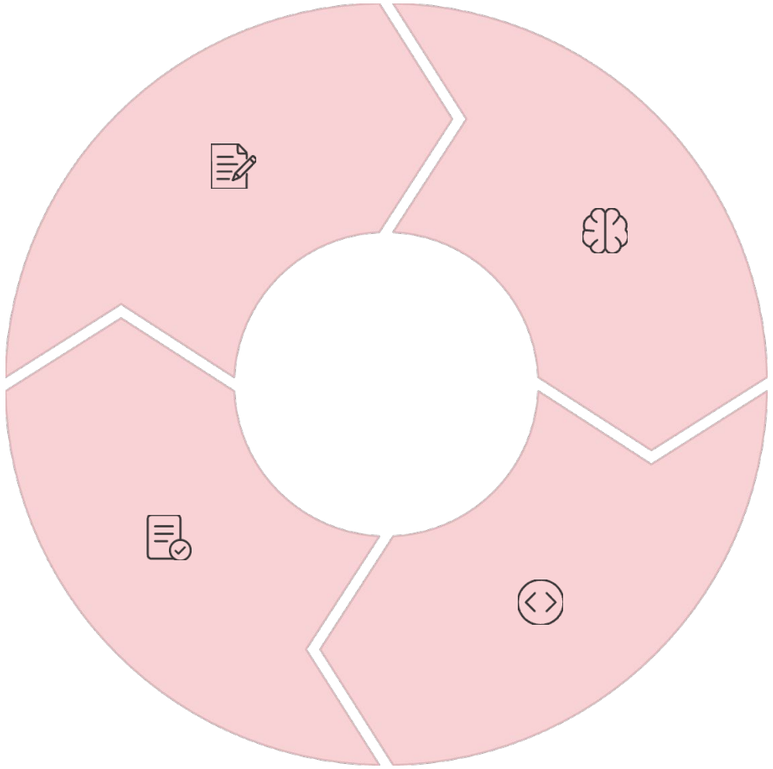
Understanding these distinctions is crucial for deploying AI effectively, whether for content creation, automated tasks, or complex problem-solving in marketing and development.

Spec-driven Development: From Requirements to Code with AI

Spec-driven Development (SDD) is a methodology where the development process is guided by a formal specification (the "spec"), acting as a single source of truth for both humans and AI agents.

1. Define & Refine Spec

Start with a technical specification



2. AI Code Generation

AI interprets the spec to scaffold boilerplate, logic and tests

3. Human Development

Developers review and integrate the AI-generated code

4. Continuous Synchronization

Ensure code evolves in sync with the spec

The easiest way to write spec are .md documents. The format of these files are - to this day - tool-dependant.
Spec topics categories: 1) CONTEXT & RULES (context information, e.g. use certain dependences). 2) SKILLS (specific tasks that we want them to perform in a certain way or things that we need to teach the Agent)

Link to GitHub repository: set your environment ready for next week

https://github.com/sbesby/ai4cm